

WORK EXPERIENCE

American Express

July 2023 – July 2024

Software Engineer - Associate DPM (Global Risk & Compliance Dept)

- **Saved ~\$70K** in Q1–Q2'24 through **33% data rationalization**, by designing & developing **Python** and **SQL** pipelines to resolve data quality issues and implementing optimization strategies for the **Hive** data layers used by the Anti-Money Laundering Transaction Monitoring team.
- Led the implementation of onboarding pipelines for Kabbage acquisition, migrating **~20K accounts** into Amex's banking ecosystem through scalable ingestion, validation, and reconciliation frameworks that ensured AML compliance.
- Maintained and debugged production pipelines ingesting **millions of transactions** per 5-minute Amex feed, detecting file loss and duplication through transaction-volume anomaly monitoring in **Yellowbrick**; partnered with the AML Transaction Monitoring team via **JIRA** in an Agile environment to triage and resolve data-integrity issues.

American Express

May 2022 – July 2022

Data Engineering - Analyst Intern (Credit & Fraud Risk Dept)

- Reduced development time by 80% for Model Performance Tracking of the Expected Credit Loss Forecasting Tool, by building **Python** scripts in **MLStudio** to consolidate all US and global portfolios.
- Automated data sourcing across multiple formats (SQL, JSON) by designing object-oriented **Java** services with secure coding principles to pull and load data into **Hive**, following SDLC.
- Streamlined onboarding for new data engineers by authoring technical design documentation in **Confluence** detailing BAU and ad-hoc workflows, sharing knowledge to support the team.

TECHNICAL SKILLS

Language: C, C++, Python, MATLAB, PHP R, SQL, PostgreSQL, HiveQL, TypeScript, JavaScript**Cloud Tech:** Amazon Web Services: EC2, S3, Lambda, ECS, CodePipeline; Google Cloud : GCP, Vertex AI, Firestore.**DevOps, Infra & Delivery:** Git, Docker, Kubernetes, Apache Kafka, Redis, Linux OS, Slurm, SDLC, CI/CD, Jenkins.**App Dev & Analytics:** HTML, CSS, REST APIs, React, Node.js, FastAPI, Uvicorn, Streamlit, n8n, PHPUnit.**ML & AI:** scikit-learn, numpy, scipy, pandas, LangChain, keras, tensorflow, LangGraph, transformers; BERT, Gemini 2, RoBERTa, all-MiniLM-L6-v2, all-mpnet-base-v2, VGG19, Opacus, Presidio, GPT-2, NetworkX.**AI Coding Tools:** Claude Code, Codex, Cursor, Github Copilot

EDUCATION

University of Michigan, Ann Arbor

Aug. 2024 – May 2026

*Master of Science in Engineering in Computer Science***GPA: 4.0/4.0****Delhi Technological University, Delhi**

July 2019 – May 2023

*Bachelor of Technology in Mathematics & Computer Science***GPA: 3.81/4.0****Coursework:** Data Structures, Algorithms, Object Oriented Programming, Software Engineering, System Design, Operating Systems, Computer Networks, Database Management Systems, Machine Learning, Natural Language Processing, Data Science, Statistics, Linear Algebra.

PUBLICATION

“**University Recommendation System**” presented at **International Research Conference on Science Technology, Engineering and Management (IRCSTEM)** on 6–7th May 2023 at Gurgaon, India.

PROJECT EXPERIENCE

Wikispeech (MediaWiki extension): Open Source Contribution

- Built Wikimedia's article-to-audio download feature, enabling users to listen to and **save Wikipedia articles as speech**; delivered 4 **PHP** patches spanning schema design, **REST APIs**, JobQueue processing, and UI workflows with 40+ **PHPUnit** tests.

CineRank: Real-Time Movie Recommendation System

- Built a real-time recommendation engine that continuously updates user preferences & serves movie suggestions in **<100ms latency** on the **MovieLens 32M** dataset. Designed a streaming ML pipeline using **Kafka**, **Redis**, **PostgreSQL**, **FastAPI**, combining SVD collab filtering, cold-start fallback logic, & cached recommendation retrieval.

Needle: RAG-based Research Assistant

- Built an RA for literature discovery & source-grounded Q&A over **1.7M+ arXiv papers** & uploaded PDFs, removing hallucinated citations through retrieval-backed generation. Developed a RAG pipeline using **Streamlit**, **Vertex AI**, **Gemini 2**, & **LangChain** with semantic search, citation verification, & conversational document querying.

PrivNLGBench: Differential Privacy Benchmark

- Built a benchmark for evaluating privacy-preserving NLP systems by transforming real-world datasets (Yelp, Enron, Clinton FOIA) into structured test sets with isolated PII and PII-free summaries. Developed the evaluation pipeline using **PyTorch**, **Opacus**, **DistilRoBERTa**, & **NetworkX**, achieving **0.85 accuracy** / **0.71 macro-F1** for PII detection and improving calibration from **0.272 to 0.083 ECE**.

MiniGPT: Built a decoder-only GPT-style **transformer from scratch** using **pytorch** & **numpy** with multi-head self-attention & positional embeddings, pretrained on web text & fine-tuned for multi-turn dialogue & MCQ answering.

AI Generated Text Detector: Built a logistic regression classifier using **scikit-learn** & **scipy** sparse features, with ~80% accuracy on AI-written text detection, benchmarking SGD, RMSprop, & AdamW optimizers in **pytorch**.

Chaos Classification on Keyword Extraction: Built a novel chaos classifier using Henon maps, Lyapunov exponents on **scikit-learn** tf-idf, **spacy** NER, **NLTK** tokenization, & word2vec (**GloVE**) features to extract chaotic keywords from a corpus.

COVID-19 Detection using CNN: Built a **python** lung X-ray classification system using **tensorflow** and **keras** with VGG19 and ResNet50 architectures, achieving 95% accuracy on COVID-19 detection.

ACHIEVEMENTS, EXTRACURRICULARS & LEADERSHIP

- **Awards:** McKinsey Next Generation Women Leaders 2023 & 2025 | URAM (Micron) Scholarship 2022 | AWS ML Scholarship 2021 | UCMAS International Distinction Grade 10
- **Leadership:** Judge, MHacks 17 (MLH Official) | Vice Chairperson, IEEE DTU CS | Student Advisory Board, Michigan Engineering | Networking Officer, GradSWE
- **Mentorship:** Mentored 300+ students at IEEE DTU, JA India & American Express, Joy of Coding (UMich)